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Prediction of soil available nitrogen using machine learning and digital mapping techniques in Northeast India

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ABSTRACT

Nitrogen (N) is an essential nutrient for crop growth, influencing both agricultural productivity and ecosystem processes. Conventional fertilizer recommendations often rely solely on soil N availability at sampling points, lacking information on its spatial variability. High-resolution spatial data are therefore crucial for sustaining sericulture production. This study represents the first effort to develop a spatial prediction map of available N in the Lakhimpur district, using 160 soil samples along with local environmental covariates. Five machine learning algorithms—Random Forest (RF), Cubist, XGBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)—were employed to predict available N in the plough layer (0–15 cm) at a 30 m resolution. Measured N values ranged from 102 to 580 kg ha^{-1}. Under random cross-validation, RF consistently outperformed other models ($R^2c = 0.96$, RMSEc = 40.69 kg ha^{-1} in calibration and $R^2v = 0.35$, RMSEv = 70.44 kg ha^{-1} in validation). In contrast, spatial cross-validation resulted in markedly lower accuracy across all models, with R^2 values varied between 0.02 and 0.27. Predicted N values varied between 172 and 438 kg ha^{-1}, while uncertainty, expressed as standard deviation, ranged from 101 to 111 kg ha^{-1}. The Dissimilarity Index (DI) and Area of Applicability (AOA) maps were used to delineate areas where model predictions are most reliable. Overall, the study demonstrates that machine learning combined with digital mapping provides a practical framework for predicting soil available N. The resulting high-resolution maps offer valuable insights for site-specific nutrient management and support precision agriculture planning in sericulture-based systems.

1. Introduction

Nitrogen (N) is one of the most essential nutrients for maintaining the environment and agricultural output. Fertilizer N-consumption is directly linked to agricultural productivity in India (Sharma et al. 2021). The N biogeochemical cycle critically affects the ecosystem functioning and climate change mitigation potential for any ago-ecosystem (Pathak and Fagodiya 2022, Fagodiya et al., 2020). The unbalanced N cycles cause water pollution, air pollution, and global warming (Kumar et al. 2020; Fagodiya et al., 2017). Generally, the spatial variation in the available soil N was not considered, and by default, N input was applied as per recommended practice. This practice induces over N-input and respective national consumption, affecting soil fertility (Lopez- Granados et al. 2004). Understanding the spatial distribution of available soil

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nitrogen is particularly critical because it provides the foundation for optimizing site-specific fertilizer inputs, evaluating carbon–nitrogen (C–N) cycle dynamics, and assessing nutrient budgets at the field, regional, and landscape scales. Accurate prediction of available nitrogen not only enhances soil health and crop production but also helps reduce non-point source pollution, a persistent environmental challenge in intensive farming systems. The distribution and availability of soil nitrogen are influenced by a combination of natural and anthropogenic drivers such as rainfall, temperature, fertilization, irrigation, tillage, land use, land cover, soil parent material, topographic features, and microbial communities (Vasu et al. 2017; Reza et al. 2017; Kumar et al. 2020). This complexity makes nitrogen prediction challenging when relying solely on conventional methods.

Traditional soil mapping and survey approaches generally fail to incorporate temporal variability. Moreover, conventional maps frequently operate at coarse spatial resolutions, limiting their relevance for precision nutrient management. In this context, there is a clear need for a systematic, cost-effective, and scientifically robust methodology to capture the spatial heterogeneity of soil nitrogen. One of the most promising approaches is Digital Soil Mapping (DSM), which integrates soil observations, environmental covariates, and computational models to generate continuous predictions of soil properties and useful for multiple disciplines, including agronomy, ecology, environmental science, land management, and soil geography (Heung et al. 2016; Van Leeuwen et al. 2022). The precision of DSM outputs surpasses that of conventional soil survey products, making it increasingly important in designing nutrient management strategies, land-use planning frameworks, and policies that minimize environmental footprints (Lagacherie et al. 2006; Taghizadeh-Mehrjardi et al. 2022). The growing role of machine learning (ML) and artificial intelligence (AI) in DSM has further enhanced its potential for predicting soil nitrogen distribution. Unlike traditional statistical models, machine learning algorithms are capable of handling non-linear, high-dimensional datasets, which are characteristic of soil–landscape interactions. Recent advances in data science, the availability of high-resolution remote sensing imagery, and improved computational power have made it possible to incorporate a wide range of covariates—such as spectral indices, terrain derivatives, climate surfaces, and biological indicators—into predictive models. This integration significantly improves the accuracy of soil nitrogen predictions, especially in heterogeneous landscapes (Mponela et al. 2020; Zhou et al. 2020; Maleki et al. 2023).

Among the machine learning models widely applied in DSM, several have demonstrated strong predictive performance for soil properties. Artificial Neural Networks (ANNs) mimic human cognition and are highly effective in capturing complex, non-linear relationships (Bahmani et al. 2020). Boosted Regression Trees (BRTs) combine decision tree algorithms with boosting techniques to enhance model robustness (Zhou et al. 2020). Cubist models are rule-based systems that partition data and apply linear regression within each partition, making them interpretable and efficient (Emadi et al. 2020; Kasraei et al. 2021). Generalized Linear Models (GLMs) and Multiple Linear Regression (MLR) remain relevant for their interpretability and computational efficiency, although their capacity to capture complex interactions is limited (Schillaci et al. 2021; Steinbuch et al. 2022).Two of the most commonly used models in DSM are Random Forest (RF) and Support Vector Machine (SVM). RF, an ensemble-based algorithm, has been shown to perform exceptionally well in modeling soil properties, including soil nitrogen, due to its ability to reduce overfitting and handle large numbers of predictor variables (Akpa et al. 2014; Jamshidi et al. 2019; Jena et al., 2023). SVM, on the other hand, is powerful in handling small datasets with high-dimensional covariates, offering reliable predictions in heterogeneous environments (Tajik et al. 2020). Collectively, these machine learning approaches have made DSM a robust methodology for soil nitrogen prediction.

Sericulture is a livelihood occupation in >22 countries globally. Silk (mulberry, muga, tasar, and eri) production not only provides livelihood support but also bears high traditional and ethnic value among the various tribal inhabitants of the Northeastern states of India (Kumar et al., 2021). Among all silk, muga silk is known for its natural golden color and is endemic to the northeastern states of India. Eri silkworm is also a food and energy source among northeast inhabitants. Assam produces >90% of the global muga and substantially contributes to eri silk production (Kumar et al., 2024). Lakhimpur has been the highest muga silk-producing district in Assam for a long time. The quality ensures the highest profitable silk production leaves production, in which soil N content plays a significant role. Silk is a protein fiber that requires a sufficient supply of N. Thus, producing qualitative healthy leaves is the prime need for adequate profitability of silk production. Existing soil N maps in northeast India face several limitations that restrict their usefulness for site-specific nutrient management. Most available maps are based on coarse-resolution datasets, which fail to capture the fine-scale variability of soil nitrogen across heterogeneous landscapes. Field validation is often limited or absent, leading to uncertainties in spatial predictions. Additionally, these maps inadequately represent the complex interactions between topography, land use, and climatic factors that influence nitrogen dynamics under diverse agroecological conditions. As a result, they offer only generalized insights, making them insufficient for precision agriculture and sustainable soil fertility management in the region. Digital soil mapping of N, powered by machine learning, provides an opportunity to operationalize precision nutrient management at scales ranging from individual farms to entire regions. Keeping this in view, the present investigation of the Lakhimpur sericultural soil for (1) determination of available soil N spatial distribution, variability, and respective influencing factors, (2) comparison and evaluation of various models' prediction performance, (3) uncertainty evaluation of the predicted output has been focused.

2. Materials and methods

2.1. Study area

Lakhimpur's favourable environmental conditions, rich soil, historical significance, and other factors have made it the world's largest district for the production of muga silk. The Lakhimpur District is located on the northeastern tip of Assam, on the north bank of the Brahmaputra River. The district is located approximately between 93°42′ and 94°20′ East longitude and between 26°48′ and 27°53′ Northern latitude (Fig. 1). Its northern boundaries are the Arunachal Pradesh districts of Siang and Papumpare; its eastern boundaries are the Subansiri River and the Dhemaji District. The Gahpur sub-division of

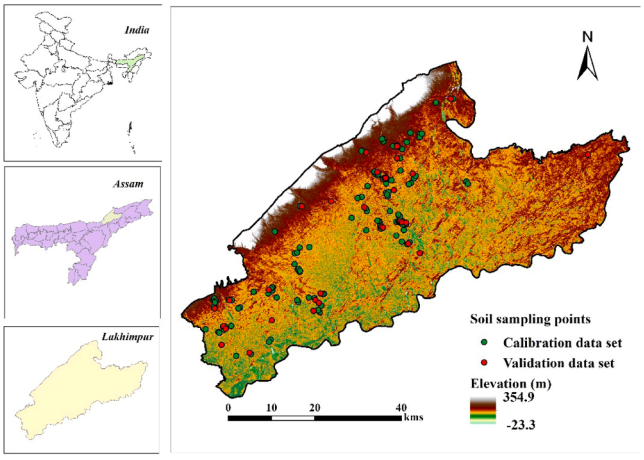


Fig. 1. Soil sampling points in the Upper Brahmaputra Valley in northeast India. Green points depict the calibration dataset, and red points represent the validation dataset.

Biswa-Nath District is on the west side, while the Brahmaputra and Majuli Districts are on the southern side. The district's centre region (2257 sq km) is classified as rural. The average yearly temperature in Lakhimpur is 24.5 °C. There has been over 2000 mm of rainfall.

2.2. Soil sampling and N analysis

Systematic randomised sampling was used to gather 160 composite samples of top soils (0–15 cm) from the Lakhimpur district of Assam. Using a v-notch technique, topsoil samples (0–15 cm) were gathered from the research area by systematic randomised sampling. The first cleaning was chosen at every test location to provide a soil sample free of debris. To capture every aspect of the field situation, 200 g of composite samples were created on the spot using the proportionate approach. Each composite sample is made up of ten individual samples. There was a 500g minimum quantity restriction on the replication. The laboratory received all of the composite replicates. After letting each duplicate air dry in the shade, they were sieved using a 2 mm sieve. Alkaline potassium permanganate was used to calculate the available N (Subbiah and Asija, 1956).

2.3. Digital soil mapping technique

The SCORPAN model uses environmental characteristics, including climate (c), age (a), geographical position (n), organisms (o), parent material (p), and relief (r), to indirectly infer soil class/properties at a given location (McBratney et al. 2003). The SCORPAN model provides the quantitative (empirical) link between soil properties and factors that regulate spatial variability. The basic concept of the SCORPAN model lies in Dokuchaev (1976) and Jenny (1941) ideas.

$$S = f(s, c, o, r, p, a, n) \quad (1)$$

The SCORPAN model may take in soil data through point data, spectral features (remotely sensed), or pre-existing soil maps. Environmental variables are digitally and geographically explicit raster data processed by a geographic information system (GIS). Quantifying the relationships between spatially explicit digital environmental factors and the expected soil qualities is facilitated by the SCORPAN model. Uncertainty estimate is more straightforward when comparing the SCORPAN model to other conventional techniques.

2.4. Environmental covariates

A total of twenty-eight environmental covariates (ECs), considered to be associated with pedogenesis, were compiled from multiple sources, including satellite-derived datasets and index products. These covariates were organized into major groups, namely climate, relief, parent material, organisms, location, remote sensing, and soil-related factors. In digital soil mapping (DSM), data from the Landsat 8 Operational Land Imager (OLI) have been extensively employed due to its 30 m spatial resolution and 16-day revisit cycle (Moharana et al. 2022). Surface reflectance products and associated vegetation indices were generated from the Landsat 8 Surface Reflectance Level-1 Tier 1 collection (USGS, accessed May 25, 2023). Among the spectral covariates, the Normalized Difference Vegetation Index (NDVI) along with reflectance data from all eleven Landsat bands were incorporated into the modeling framework to aid in predicting soil nitrogen (Table 1). Relief-based covariates were computed using a 30 m digital elevation model (DEM) derived from the Shuttle Radar Topography Mission (SRTM). Various terrain attributes were generated in SAGA-GIS version 6.3.0, including channel network base level (CNBL), convergence index (CI), channel network distance (CND), multi-resolution index of valley bottom flatness (MRVBF), LS factor, plan curvature (PlaC), profile curvature (PrC), total catchment area (TCA), topographic position index (TPI), topographic wetness index (TWI), topographic roughness index (TRI), and valley depth (VD).Climatic variables such as mean annual temperature (MAT) and mean annual precipitation (AP) were extracted from the WorldClim version 2 database (http://www.worldclim.com/version2, accessed May 25, 2023). To harmonize the spatial datasets, all covariates were resampled to a consistent 30 × 30 m grid, using average re-sampling for coarser data and bilinear re-sampling for finer-scale data. Each variable was prepared in raster format and geo-referenced to the Universal Transverse Mercator (UTM) coordinate system, WGS84 datum, Zone 46N. These harmonized covariates provided the foundation for constructing

Table 1
Various environmental covariates used in the models.

Environmental covariates	Abbreviation	Resolution	Mean	Min	Max
Aspect	Asp	30m	3.18	0.00	6.28
Convergence Index	CI	30m	5.39	-47.18	51.21
Channel network base level	CNBL	30m	82.43	70.75	109.97
Channel network distance	CND	30m	7.03	0.00	22.51
Elevation (m)	Elev	30m	88.53	59.67	116.40
LS-factor	LS_Factor	30m	2.67	0.02	9.90
Multi-resolution Ridge Top Flatness Relative slope position	MRRTF	30m	0.65	0.00	4.42
Multi-resolution Index of Valley Bottom Flatness	MRVBF	30m	0.73	0.00	5.80
Relative slope position	RSP	30m	0.11	0.00	0.43
Slope	Slp	30m	0.10	0.00	0.26
Total catchment area	TCA	30m	78638	959	4892340
Topographic Positioning Index	TPI	30m	0.85	-11.19	10.49
Topographic Roughness Index	TRI	30m	3.23	1.08	8.40
Valley depth	VD	30m	59.45	18.95	100.15
Landsat Band1	Band1	30m	11112.31	10693.90	14115.70
Landsat Band2	Band2	30m	10121.27	9613.74	13974.50
Landsat Band3	Band3	30m	9615.18	8952.72	14430.60
Landsat Band4	Band4	30m	8488.40	7539.60	15172.70
Landsat Band5	Band5	30m	18392.15	14978.70	21984.60
Landsat Band6	Band6	30m	13582.21	11266.50	20536.80
Landsat Band7	Band7	30m	9506.82	7657.73	18707.30
Landsat Band8	Band8	30m	9091.32	8331.01	14683.00
Landsat Band9	Band9	30m	5048.36	5024.32	5068.92
Landsat Band10	Band10	30m	28404.28	27930.40	30232.70
Landsat Band11	Band11	30m	25195.66	24840.80	26183.80
Normalized Difference Vegetation Index	NDVI	250 m _16 days	0.37	0.08	0.46
Annual Precipitation	AP	1 km	3019.87	2554.71	3253.06
Annual Mean Temperature	MAT	1 km	23.98	23.65	24.15

quantitative spatial prediction models.

2.5. Machine learning techniques (prediction models)

In this study, five machine learning (ML) algorithms were selected for evaluation, namely Random Forest (RF), Support Vector Machine (SVM), Extreme Gradient Boosting (XGBoost), Cubist regression tree, and K-nearest Neighbor (KNN). Their inclusion was guided by demonstrated predictive performance, methodological suitability, and supporting evidence from previous applications reported in the scientific literature.

2.5.1. Random forest

The Random Forest (RF) algorithm operates on the principle of binary recursive partitioning and is constructed from an ensemble of Classification and Regression Trees (CART). Each tree in the ensemble is generated by repeatedly splitting the dataset into nodes based on predictor variables, thereby establishing relationships between explanatory and response variables. By aggregating the predictions of multiple trees, RF enhances model accuracy and reduces overfitting compared to single decision trees. Two parameters are particularly important in optimizing model performance: the number of trees to be grown in the forest (Ntree) and the number of predictor variables randomly sampled at each node split (Mtry). These parameters can be systematically tuned during the modeling process to achieve more robust and reliable predictive outcomes. The aggregate of the all-individual tree's (algorithms) outcomes is the ultimate forecast made by RF. Bootstrapping the original datasets achieves an Out-of-bag (OOB) estimation of error. When all of the trees' OOB forecasts are combined, the result is Mean Square Error (MSEOOB) (Liaw and Wiener, 2002):

$$\mathrm{MSE_{OOB}} = \frac{1}{n}\sum_{i=1}^{n}\left(z_i - \widehat{z}_i^{OOB}\right)^2 \tag{2}$$

where $\widehat{\boldsymbol{z}}_{\boldsymbol{i}}^{\boldsymbol{OOB}}$ is the average of all OOB predictions. The percentage of explained variance ($\mathrm{Var_{ex}}$) is determined after the $\mathrm{MSE_{OOB}}$ is normalized due to its dependency on the response variable unit:

$$\mathrm{Var_{ex}} = 1 - \frac{\mathrm{MSE_{OOB}}}{\mathrm{Var_z}} \tag{3}$$

where $\mathrm{Var_z}$ denotes the answer variable's overall variance.

For this reason, RF is useful in predicting the output of ecological variables in the context of composite environmental systems.

2.5.2. Cubist

The Cubist algorithm is essentially built upon linear regression principles, where regression models form the terminal nodes of regression trees (Taghizadeh et al. 2021). In recent years, digital soil mapping (DSM) studies have demonstrated the growing importance of such machine learning methods across diverse ecological and geomorphic settings (Emadi et al. 2020; Mahmoudzadeh et al. 2020). Within the Cubist framework, linear regression functions are embedded within the decision-tree structure, producing multivariate relationships that are expressed as sets of rules (Fan et al. 2018). The final predictive outcome is generated by selecting and combining these rules, which provide flexibility and accuracy across complex datasets. Two key parameters influence model behaviour: the number of committees (ensembles of model trees), which may range between 1 and 100, and the number of nearest neighbours incorporated into predictions, typically set between 0 and 9. This rule-based regression tree environment allows Cubist to capture both linear and non-linear patterns within soil–environmental covariate relationships. For the present study, model development and prediction were implemented using the Cubist package in the R statistical environment.

2.5.3. Extreme gradient boosting

Extreme Gradient Boosting (XGBoost) is widely applied to enhance the predictive accuracy of decision tree–based classification and regression models. By introducing regularization and optimizing objective functions, XGBoost reduces the risk of overfitting while improving computational efficiency. A key advantage of this algorithm lies in its ability to perform parallelized learning, allowing multiple functions to be updated simultaneously during the training process. XGBoost employs the principle of boosting, where several weak learners are combined iteratively to generate a strong predictive model, thereby improving overall robustness and stability (Fan et al., 2018). For model calibration, several hyperparameters were carefully tuned to balance complexity and generalization.

2.5.4. Support vector machines

SVMs may predict each multivariate function with a given degree of accuracy. SVMs resolve regression issues with both linear and nonlinear models. The nonlinear model expands the input space, while the linear model finds noise in the datasets (Hou and Behdinan 2022). The SVM classification and regression process applied interconnected supervised learning algorithms. These methods are very accurate univariate predictors of multivariate functions. Various components, including input data, kernel type function, penalty and bandwidth factors, were used in the current investigation.

2.5.5. K-nearest neighbor

In the ongoing DSM investigation, the KNN (nonparametric approach) algorithm is employed for regression analysis and classification. Using a distance metric, this technique finds the k data points in the training set to categorise a new data point (Hastie et al., 2009).

2.6. Model evaluation

In this study, four statistical measures were applied to evaluate model performance in random cross-validation: root mean square error (RMSE), mean error (ME), coefficient of determination (R^2), and Lin's concordance correlation coefficient (CCC). The RMSE and R^2 values were particularly emphasized for assessing model accuracy and stability. A lower RMSE indicates improved predictive accuracy, whereas a higher R^2 reflects stronger model stability. Model calibration and validation were carried out using an 80:20 split of the dataset. The predictive ability of the models was further examined by comparing RMSE values from calibration (RMSEc) and validation (RMSEv) phases, which provided insights into both fitting accuracy and generalization capability.

The following was used to estimate the validation metrics:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(p_i - o_i)^2}{\sum_{i=1}^{n}(\widehat{p}_i - \widehat{o}_i)^2} \tag{4}$$

$$\mathrm{ME} = \frac{1}{n}\sum_{i=1}^{n}(p_i - o_i) \tag{5}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(p_i - o_i)^2} \tag{6}$$

where, $\boldsymbol{p_i}$ and $\boldsymbol{o_i}$ are predicted and observed values, $\widehat{\boldsymbol{p}}_{\boldsymbol{i}}$and $\widehat{\boldsymbol{o}}_{\boldsymbol{i}}$are means of predicted and observed values

$$\mathrm{CCC} = \frac{2\rho\sigma_0\sigma_p}{{\sigma_0}^2 + {\sigma_p}^2 + \left(\mu_0 - \mu_p\right)^2} \tag{7}$$

$\boldsymbol{\mu}_0$and$\boldsymbol{\mu}_{\mathbf{p}}$are the means of observed and predicted values, and ${\boldsymbol{\sigma}_0}^2$ and ${\boldsymbol{\sigma}_{\boldsymbol{p}}}^2$ are corresponding variances ρ is the Pearson correlation coefficient between observed and predicted values

In addition to random cross-validation, spatial or block cross-validation was applied using the blockCV package in R to evaluate model performance while accounting for spatial autocorrelation in the dataset. Unlike random cross-validation, which may overestimate predictive accuracy due to spatial clustering, block cross-validation partitions the study area into spatially distinct blocks. Training and testing subsets were then generated by systematically excluding entire blocks during model fitting. This ensured that validation was conducted on spatially independent data, providing a more realistic measure of model transferability. The use of the blockCV package has been widely recommended in soil studies to reduce bias and improve the reliability of spatial prediction models.

The Area of Applicability (AOA) concept was applied to assess this source of uncertainty. The procedure includes two main steps: (a) calculating the Dissimilarity Index (DI), which measures the minimum distance of each prediction point from the training data within the multivariate predictor space, and (b) defining the 0.95^{th} quantile threshold of the DI from the training data, based on multiple DI values generated through cross-validation. Both AOA and DI were computed using the *aoa* function available in the R package CAST. The resulting AOA analysis generates a binary map, where a value of '1' indicates regions within the AOA, signifying reliable model predictions, and a value of '0' marks areas outside the AOA, indicating uncertain predictions.

To assess spatial autocorrelation, experimental semivariogram was used to examine the variance between pairs of data over a varying range of distances. The R package geoR was used to examine the semivariogram of the residuals.

To evaluate the contribution of individual observations to the overall dataset and to quantify predictive uncertainty, the Prediction Interval Coverage Probability (PICP) was employed as a key criterion. PICP measures the proportion of observed values that fall within a specified prediction interval, providing a robust statistical assessment of model reliability. By constraining PICP to prediction intervals corresponding to defined confidence levels, it is possible to estimate the uncertainties associated with model forecasts and assess the degree to which the model captures the variability in the data (Lagacherie et al., 2019; Dharumarajan et al., 2021). In this study, confidence intervals of 99, 97.5, 95, 90, 80, 60, 40, 20, 10, and 5% were considered, covering a wide spectrum from highly conservative to more permissive intervals. For practical applications, the observed PICP is expected to closely approximate the nominal coverage of its corresponding confidence interval. For example, a 90% prediction interval ideally encompasses approximately 90% of observed values, reflecting minimal prediction uncertainty, while the lower bound of uncertainty is generally near 100 $(1 - \alpha)$% for a given confidence level (Solomatine and Shrestha, 2009; Malone et al., 2011). This approach allows for a clear interpretation of model reliability and highlights areas where predictions may be less certain. In digital soil mapping and related geospatial modeling, such uncertainty quantification is essential for informed decision-making, as it helps to identify regions of high prediction confidence and areas where additional sampling or model refinement may be required. By systematically applying PICP across multiple confidence levels, the study provides a comprehensive framework for evaluating both model performance and uncertainty in spatial predictions.

2.7. Uncertainty assessment

The accuracy of a digital soil map is often evaluated at the specific locations where soil samples are collected and analyzed, providing a measure of model performance at those points. However, site-based validation does not capture the spatial variability of predictions across the entire mapped area. Assessing uncertainty at the level of individual pixels or raster cells offers a more comprehensive understanding of model reliability. In the present study, the standard deviation of predicted soil properties across 50 model runs was computed for each pixel to quantify the variation in predictions. Pixel-level uncertainty is particularly important in digital soil mapping, as it identifies areas where predictions are less reliable, guiding sampling strategies, model refinement, and informed decision-making for soil management and precision agriculture (Jena et al., 2023).

2.8. Software

Data pre-processing, digital mapping, and statistical analyses were conducted using the R programming environment (R Core Team, 2024), while map production and visualization were performed in ArcGIS software.

3. Results and discussion

3.1. Descriptive statistics

The dataset consisting of 160 soil samples was divided into calibration and validation subsets, with 128 samples (80%) allocated for model calibration and 32 samples (20%) reserved for validation. Descriptive statistical analysis revealed that the calibration set closely reflected the overall dataset, with comparable values of kurtosis, skewness, and coefficient of variation (Table 2). The observed available nitrogen (N) concentrations ranged from 102 to 580 kg ha^{-1}, indicating considerable variability across the study area. This wide range can be attributed to the region's annual flooding events, which strongly influence the soil's nutrient dynamics. The upper Brahmaputra valley's surface soils appear highly sensitive to environmental influences, management practices, and disturbances caused by both natural and anthropogenic factors. The relatively high cumulative variation of 31.7% highlights that available N exhibits a semi-homogeneous distribution, with notable spatial heterogeneity. Such variability is likely driven by factors including land use changes, differences in crop management practices, variations in topography, and localized environmental conditions. Similar patterns of spatial variability in soil N have been reported in other studies across North-eastern India, emphasizing the combined effects of natural and human-induced processes (Sheldon et al., 2018; Jena et al., 2022). These findings reinforce the need for modern geostatistical and machine learning approaches to capture the underlying spatial structure of available N, rather than relying solely on mean values or point measurements. Available N content can be considered a stochastic variable influenced by multiple interacting factors. Inputs such as synthetic fertilizers, organic amendments including animal manure, leaf litter deposition, flooding regimes, and localized climate variations collectively determine nutrient availability at a given site (Moharana et al., 2017). Recognizing these interactions is crucial for designing site-specific nutrient management strategies that optimize fertilizer application, reduce environmental loss, and support sustainable agricultural production. Consequently, high-resolution predictive models that incorporate both soil and environmental covariates are essential for accurately mapping the spatial distribution of available N, enabling informed decision-making for precision agriculture in flood-prone and heterogeneous landscapes.

Rug plots along the X-axis indicate the sampling locations and corresponding available N contents (Fig. 2), while the overlaid bell-shaped curve illustrates the overall normal distribution of N values. Since machine learning algorithms do not require normally distributed input data, no normalization was applied, and the models were fitted directly using the measured N concentrations from the study sites. Probability density curves and histogram bars visually represent the frequency distribution of N content across the landscape. To assess the spatial structure and potential clustering of N values, Global Moran's I was computed. The results indicated no significant spatial autocorrelation, confirming that the distribution of available N was sufficiently random for unbiased model training. Ensuring the absence of spatial dependence is critical because unaccounted autocorrelation can artificially inflate

Table 2
Descriptive statistics of soil available N content in the Upper Brahmaputra Valley study area in northeast India.

Data set	N	Mean	SE	Median	SD	CV	Skewness	Kurtosis	Min	Max
Total	160	313	7.85	305	99	31.7	-0.035	-0.330	102	580
Calibration	128	317	8.73	319	99	31.2	-0.25	-0.47	102	503
Validation	32	299	18	271	101	34	0.84	1.11	119	580

SE, standard error; SD, standard deviation; CV, coefficient of variation.

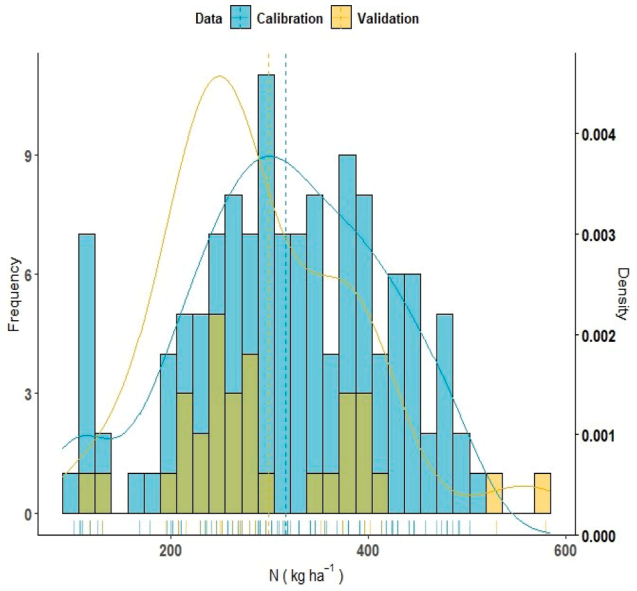


Fig. 2. The distribution of soil available N in the calibration and validation dataset is represented by a density plot, histogram, and rug plot. The mean is shown as vertical dashed lines.

model performance. This approach supports the reliable application of machine learning for predicting soil N, allowing high-resolution mapping while maintaining the statistical independence of sampling points, a key requirement for robust predictive modeling.

3.2. *Hyperparameters of machine learning models*

Table 3 summarizes the hyperparameters configured for each machine learning model applied in the prediction of available N. These parameters were carefully selected and optimized to ensure model efficiency, accuracy, and stability during both calibration and validation phases.Fig. 3 illustrates the RF hyperparameter optimization process for N prediction, where the model performance was evaluated across different combinations of ntree and mtry. The results show that RMSE varied notably with changes in both ntree and mtry. At lower mtry values (2–4), the RMSE values were consistently lower across all tree settings, indicating more stable and accurate predictions. As mtry increased beyond 6, RMSE tended to rise, suggesting that including too many variables at each split introduced noise and reduced generalization performance. Among the tested configurations, the model achieved the lowest RMSE (~99.5) when ntree = 400 and mtry = 6, demonstrating that a moderate number of trees and limited variable sampling yield optimal performance. Additionally, increasing the number of trees beyond 400 did not significantly improve accuracy. Instead, RMSE slightly increased for larger ensembles (ntree = 800 and 1000), implying diminishing returns with higher tree counts and longer computation times. The optimization results highlight that the most effective RF configuration for predicting available N was achieved at ntree = 400 and mtry = 6, balancing computational efficiency with predictive reliability. This tuned parameter combination was subsequently adopted for the

Table 3
Hyperparameters of machine learning models used for available N prediction.

Model	Hyperparameters	Description
RF	ntree = 400	Number of trees grown in the forest
	mtry = 6	Number of variables randomly sampled at each split
	nodesize = 5	Minimum size of terminal nodes
	importance = TRUE	Enables variable importance estimation
Cubist	committees = 50	Number of model committees to improve accuracy
	neighbors = 5	Number of nearest neighbors used for rule adjustment
	rules = auto	Automatically determined based on data complexity
SVM	kernel = radial	Radial basis function (RBF) kernel used
	C = 1	Regularization parameter controlling margin width
	epsilon = 0.1	Insensitive loss function parameter for regression
	gamma = 0.05	Kernel coefficient defining influence of single training points
XGBoost	nrounds = 500	Number of boosting iterations
	eta = 0.1	Learning rate controlling the contribution of each tree
	max_depth = 6	Maximum depth of individual trees
	subsample = 0.8	Fraction of samples used per tree (to prevent overfitting)
	colsample_bytree = 0.8	Fraction of predictors randomly sampled per tree
	min_child_weight = 1	Minimum sum of instance weight needed in a child node
KNN	k = 5	Optimal number of neighbors determined by cross-validation
	distance = Euclidean	Distance metric used for neighbor computation
	scale = TRUE	Data normalized to equalize variable influence

RF, Random Forest; SVM, Support Vector Machine; XGBoost, Extreme Gradient Boosting; KNN, k-Nearest Neighbors

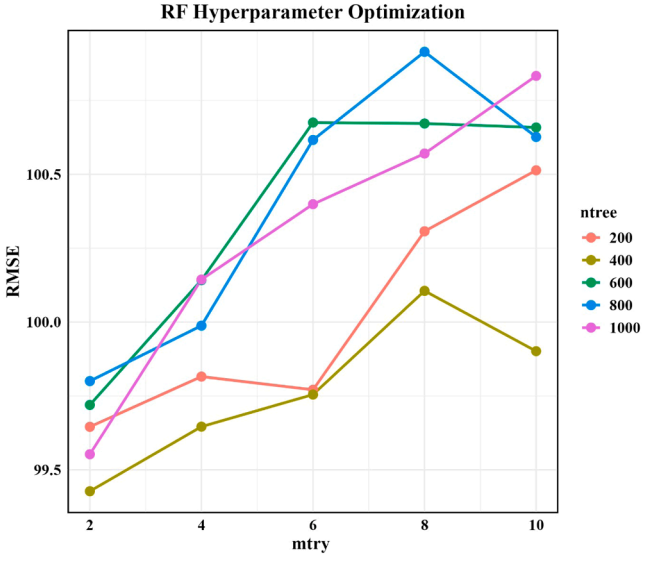


Fig. 3. Optimization of Random Forest hyperparameters (mtry and ntree) for available N prediction.

final RF model used in the study.

The Cubist model was configured with committees = 50, which increased model robustness by combining multiple rule-based committees. For the SVM model, the radial basis function (kernel = radial) was used due to its effectiveness in capturing nonlinear soil–environment interactions. The penalty parameter (C = 1) controlled the trade-off between margin width and misclassification, while epsilon = 0.1 defined the insensitive zone for regression errors. The kernel coefficient (gamma = 0.05) determined the influence range of training samples, ensuring smoother decision boundaries. The XGBoost model, known for its gradient-boosting framework, employed nrounds = 500 to define the number of boosting iterations and a learning rate of eta = 0.1 to gradually update tree weights, reducing overfitting risk. Tree complexity was managed with a max_depth = 6, while subsampling parameters (subsample = 0.8 and colsample_bytree = 0.8) maintained randomness in feature and sample selection to enhance generalization. The min_child_weight = 1 parameter further controlled the minimum instance weight required in leaf nodes to prevent overly specific fits. The KNN algorithm was implemented with k = 5, determined through cross-validation as the optimal balance between bias and variance. These parameter settings reflect the methodological diversity of the models and their tailored tuning to effectively capture the complex relationships between soil properties, terrain variables, and remotely sensed covariates influencing available N distribution.

3.3. Evaluation of prediction models

The performance of five machine learning models—RF, Cubist, SVM, XGBoost, and KNN—was evaluated using both random and spatial cross-validation techniques to predict available soil N content (Table 4). Under random cross-validation (80:20 split), the RF model exhibited the highest calibration accuracy with an $R_c^2$of 0.96 and concordance correlation coefficient (CCC_c) of 0.87, coupled with a low RMSE of 40.6. However, during validation, the performance decreased substantially (R_v^2 = 0.35, CCC_v = 0.38), indicating possible overfitting to the calibration data. Cubist and SVM showed moderate-to-low predictive strength during random cross-validation, with calibration R_c^2values of 0.39 and 0.16, respectively. Both models experienced a considerable drop in validation performance, implying weak generalization. The high RMSE values for Cubist (330.8) and SVM (96.0) suggest instability when applied to independent datasets.XGBoost achieved a very high calibration performance (R_c^2 = 0.97, CCC_c = 0.98) with minimal RMSE (23.8), reflecting strong model fitting. However, a sharp decline was observed during validation (R_v^2 = 0.12, RMSE = 103.3), further confirming its sensitivity to spatial heterogeneity. Similarly, the KNN model recorded low accuracy across all stages, with validation R_v^2 = 0.09and high RMSE (84.0), showing limited adaptability to complex soil–environment relationships. The high validation R^2 and low error indicate that RF is well-suited for capturing the spatial heterogeneity of available N in sericultural soils of the region, consistent with its recognized strength in handling nonlinear relationships and high-dimensional datasets (Dharumarajan et al., 2017; Pahlavan-Rad and Akbarimoghaddam, 2018). These results reflect its limited ability to capture complex spatial patterns of nitrogen distribution in the study area, likely due to its reliance on local neighborhood averaging, which may not fully exploit the influence of topographic, spectral, and climatic covariates. The evaluation highlights the RF model as the most reliable approach for predicting available soil nitrogen in the sericultural soils of northeast India. Its superior performance can be attributed to its ensemble structure, which reduces overfitting while effectively capturing nonlinear relationships among multiple environmental covariates. While XGBoost and Cubist performed well in calibration, their lower validation accuracies underscore the importance of independent testing to avoid overestimating model performance. The findings demonstrate that machine learning models, particularly RF, can generate high-resolution spatial predictions of soil nitrogen, providing critical information for site-specific nutrient management.

In comparison with random cross-validation, spatial cross-validation using a 5-fold spatial blocking approach provided deeper insights into model generalizability under spatial dependency constraints (Fig. 4). Each subplot represents one-fold, where points correspond to test data and training data. Unlike random partitioning, this spatially explicit method reduces the influence of spatial autocorrelation by ensuring that neighboring points are not simultaneously used for training and testing. Such a strategy provides a more conservative yet robust assessment of model performance, reflecting its true predictive ability under realistic spatial prediction scenarios. The results revealed a notable reduction in performance for all models compared to random cross-validation, confirming spatial autocorrelation effects. The RF model remained the best performer with R^2 = 0.27 and CCC = 0.30, though RMSE increased to 102.1, showing reduced accuracy in predicting spatially distinct samples. XGBoost, Cubist, SVM, and KNN exhibited poor spatial transferability, with R^2values below 0.10 and RMSEs exceeding 100, reflecting their limited capability to capture spatial variation in soil available N. The RF model demonstrated the most balanced performance across all validation strategies, combining relatively higher accuracy and stability. However, the noticeable decline in metrics under spatial blocking emphasizes the importance of incorporating spatial structure in model evaluation. These results underline the necessity for spatially explicit model validation when applying machine learning approaches to soil property prediction, ensuring realistic and transferable outcomes for landscape-scale applications.

Fig. 5 illustrates the spatial analysis of residuals derived from the RF model, representing the differences between predicted and observed N values. The empirical variogram displays the semivariance, defined as the squared difference between residuals plotted against the distance separating sample locations. According to Tobler's first law of geography, nearby observations are expected to be more similar, implying that semivariance should increase with distance before stabilizing. Key features of interest include the sill, where the semivariance levels off, and the range, indicating the spatial extent of correlation. However, in this study, the N residuals showed minimal spatial dependence. Across all distance lags, the semivariance values exhibited broad and overlapping confidence intervals, suggesting weak or negligible spatial structure. The nearly flat shape of the semivariogram confirms that the residuals

Table 4
Performance of various models for predicting available N in the Upper Brahmaputra Valley study area in northeast India.

Model	Random cross-validation (80:20)								Spatial cross-validation			
	Calibration				Validation				5-fold spatial blocking			
	R_c^2	CCC_c	$RMSE_c$	ME_c	R_v^2	CCC_v	$RMSE_v$	ME_v	R^2	CCC	RMSE	ME
RF	0.96	0.87	40.6	-0.39	0.35	0.38	70.4	3.05	0.27	0. 30	102.1	-1.16
Cubist	0.39	0.40	330.8	-315.9	0.10	0.15	137.7	32.3	0.09	0.10	108.2	-5.76
SVM	0.16	0.20	96.0	4.03	0.13	0.14	103.4	21.5	0.04	0.09	104.2	0.27
XGBoost	0.97	0.98	23.8	0.07	0.12	0.13	103.3	5.3	0.03	0.05	108.7	-1.69
KNN	0.11	0.14	94.0	-11.0	0.09	0.11	84.0	8.1	0.02	0.03	108.9	5.24

R^2: Coefficient of determination, CCC: Lin's concordance correlation coefficient; RMSE: Root mean squared error, ME: Mean error.

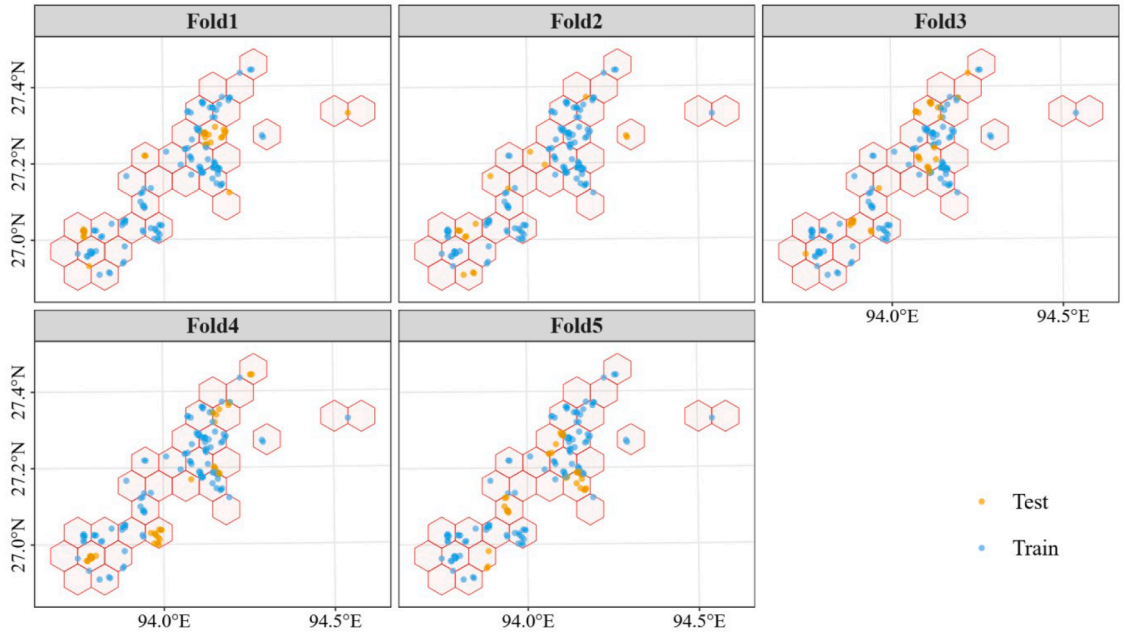


Fig. 4. Spatial block cross-validation folds for nitrogen modelling in the study area.

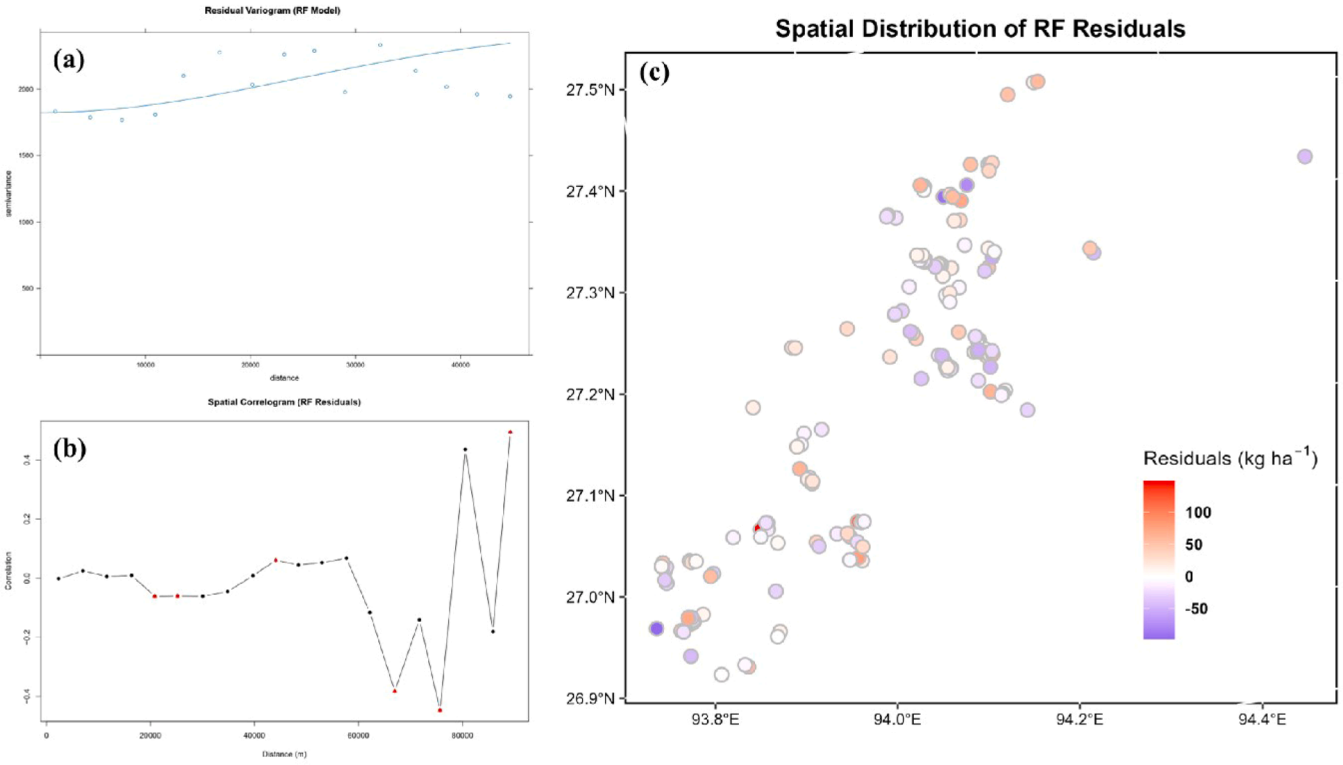


Fig. 5. Spatial analysis of RF residuals: (a) Variogram of residuals, (b) Spatial correlation, (c) Spatial distribution of RF residuals.

possess limited spatial autocorrelation. Moran I statistic indicated that value was slightly negative (-0.023), very close to zero. This indicates almost no spatial autocorrelation in the residuals and the residuals are spatially random. This indicates that the RF model effectively captured the dominant spatial variability of soil N, leaving only random noise unexplained. Consequently, the residuals' spatial independence validates the robustness and reliability of the RF model in accounting for spatially structured variation in N distribution across the study landscape.

Pearson correlation coefficient (r-value) between the predicted values by RF and XGBoost was found to be the highest (0.98) by comparing the correlation and statistical distribution (Fig. 6a). Cubist and XGBoost show moderate correlations, while KNN has the lowest, suggesting more diverse predictions. The highest discrepancy was observed in the output of the KNN model, which was verified by the Taylor diagram (Fig. 6b). Taylor's diagram summarises the variance and agreement between the observed and expected values. The Taylor diagram presented summarizes the comparative performance of five

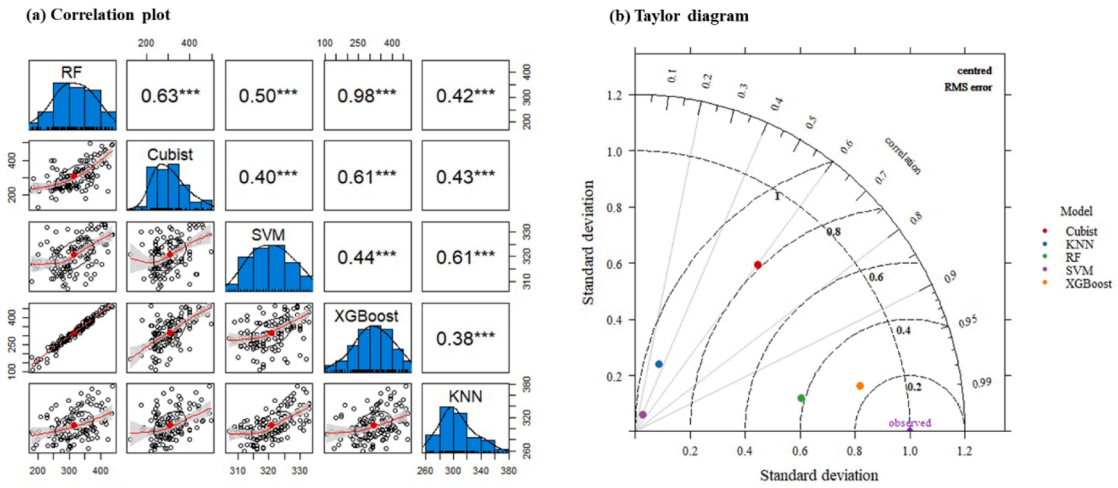


Fig. 6. Comparisons of correlations among DSM models and their statistical distributions (a) and Taylor diagram of DSM models (b).

machine learning models in predicting available soil nitrogen. The diagram simultaneously illustrates the standard deviation, correlation coefficient, and root mean square (RMS) error of each model relative to the observed values, providing a concise visual assessment of model accuracy and variability (Fig. 6b).In the diagram, the observed data are represented as a reference point with a standard deviation of 1.0, correlation of 1.0, and zero RMS error. The RF model is positioned closest to the observed reference, indicating high correlation and low RMS error, with standard deviation closely matching the observed variability. This confirms that RF effectively captures the spatial variability of nitrogen content. XGBoost shows slightly lower correlation and moderate standard deviation, reflecting overfitting during calibration and reduced generalization in validation. Cubist exhibits moderate correlation with higher RMS error and standard deviation, suggesting partial capture of variability but lower predictive accuracy. SVM lies closer to the origin with moderate correlation and low standard deviation, indicating underestimation of variability in the predictions. KNN is farthest from the observed point, with low correlation and standard deviation, confirming its poor performance in reproducing both the magnitude and variability of soil nitrogen. The Taylor diagram confirms that among the tested models, RF provides the most reliable predictions, closely aligning with observed soil nitrogen patterns, while KNN performs the least accurately.

The expected performance of uncertainties during testing is evaluated using the PICP approach. The PICP is the proportion of observations within the related prediction interval. This may be achieved by assessing the prediction intervals around different observed confidence levels. It is clear what to expect when the confidence level and coverage probability are closely watched along the 1:1 line, as the graphs in Fig. 7 demonstrate. The probability above the 1:1 line, according to (Jena et al. 2022) somewhat overestimated the extent of uncertainty. Uncertainty was discovered in the lower and upper expected boundaries with a 90% confidence range. Using the PICP approach, the efficacy of the anticipated uncertainties during testing has been assessed. The uncertainty analysis and the ML algorithms' ability to predict N showed similar tendencies.

3.4. *Importance of environmental variables*

The role of environmental variables in digital soil mapping (DSM) is crucial, as these factors govern the spatial distribution of soil properties and nutrient availability. In the present study, relative variable importance was assessed across four machine learning (ML) models to identify the key predictors of soil available N. The relative significance plots (Fig. 8) illustrate differences in variable selection and ranking among these models, highlighting how specific environmental covariates contribute to accurate N prediction.For the RF model, spectral variables such as Band 7, Band 1, and Band 4 were the most influential, followed by terrain derivatives including slope (Slp) and elevation (Elev). This dominance of spectral bands suggests that remote sensing data captured surface reflectance variations closely linked to soil depth. The moderate contribution of topographic indices (e.g., TPI and TRI) indicates that terrain-driven redistribution processes also play a key role in soil formation across the semiarid landscape.The SVM model showed a different pattern, emphasizing morphometric terrain attributes such as multi-resolution ridge top flatness (MRRTF), terrain ruggedness index (TRI), and annual precipitation (AP) as the top predictors. Climatic (MAT, AP) and terrain-based (LS_Factor, Asp) variables were more significant than spectral ones, suggesting that SVM captured broader environmental gradients rather than localized surface variability. In the XGBoost, the top predictors included TRI, Band 5, Band 10, and LS_Factor, along with elevation and TPI. These variables reflect both spectral and topographic controls, reinforcing that soil N is influenced by landscape position and surface reflectance properties. For the KNN, similar to SVM, MRRTF, TRI, AP, and MAT emerged as key predictors, followed by LS_Factor and Asp. RF evaluates the relative importance of all covariates without excluding predictors, thus preserving potential correlations between soil properties and environmental variables (Akpa et al., 2014). This difference allows RF to better capture complex nonlinear relationships and interactions among terrain, climate, and spectral factors. Similar observations have been reported in other regions; for instance, Pahlavan-Rad and Akbarimoghaddam (2018) found that elevation and CNBL were the most influential factors for soil property prediction in Zahak County, Iran. Topographic attributes, in particular, emerged as dominant factors for N prediction. TRI, MRRTF, and LS-factor reflect processes such as accumulation, deposition, erosion, leaching, and redistribution, all of which shape nutrient availability. For example, total catchment area (TCA) not only indicates water accumulation but also serves as a proxy for other soil properties influenced by hydrological processes (Dharumarajan et al., 2017; Moharana et al., 2022). Similarly, the multi-resolution index of valley bottom flatness (MRVBF) identifies depositional areas where water, salts, and organic matter are likely to accumulate, making it a useful predictor of nutrient concentration (Gallant and Dowling, 2003). These findings align with previous research demonstrating the critical role of terrain-derived covariates in ML-based N modeling (Hengl et al., 2015; Lagacherie et al., 2019).Vegetation indices, such as NDVI and EVI, are often used as proxies for soil nutrient availability because plant growth responds directly to nitrogen content, thus linking surface reflectance patterns to soil fertility (Da Silva Chagas et al., 2016). The present study

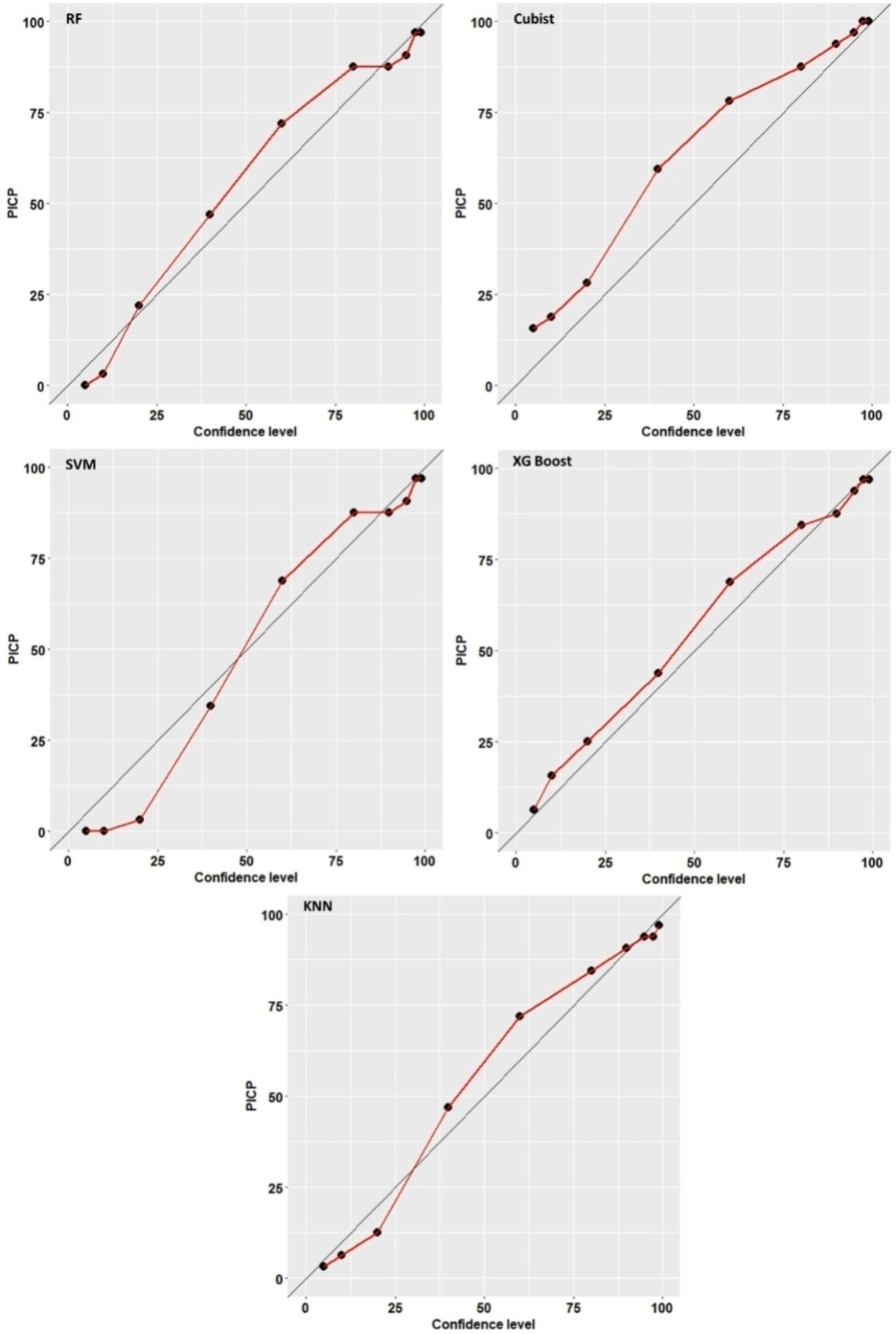


Fig. 7. Validation of the five selected models for available N between confidence level (CI) and prediction interval coverage probability (PICP).

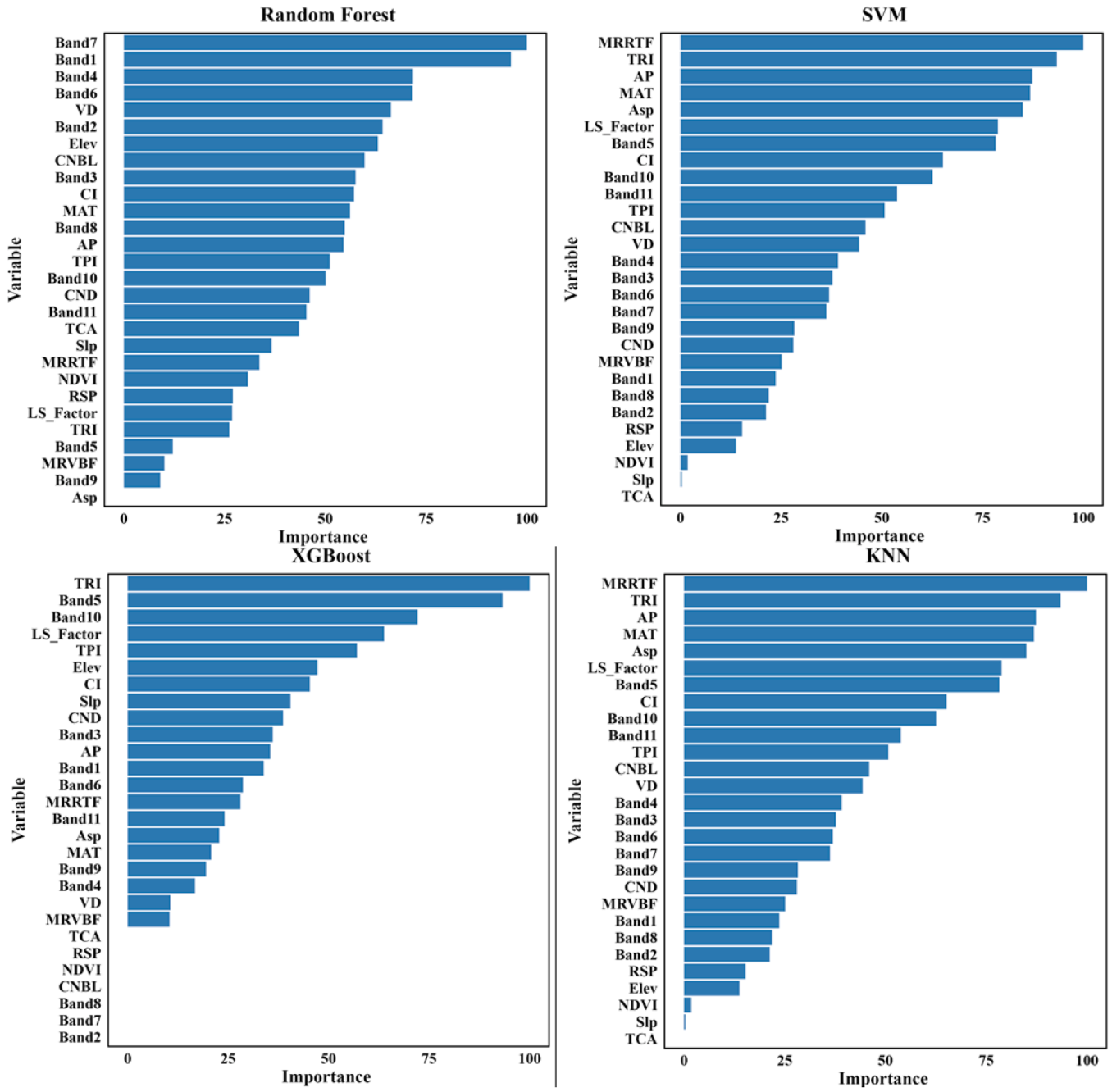


Fig. 8. Important variables in the prediction of soil availableN by various models.

corroborates these findings, as spectral information consistently contributed to model performance across multiple algorithms. Climatic variables, though secondary to topography and spectral data in importance, also played a role. Mean annual temperature (MAT) influenced RF model predictions, suggesting that temperature-mediated processes, such as organic matter mineralization and microbial activity, may affect N availability. Additionally, relative moisture conditions, particularly in the upper Brahmaputra valley of northeast India, modulate the significance of other environmental predictors (Mishra et al., 2020). Areas with fluctuating moisture levels exhibit variable N mineralization and leaching, emphasizing the need to include climatic covariates in predictive models.The use of multiple covariates in combination protects ML models from losing important predictors, as removal of key variables can degrade prediction accuracy (Prasad et al., 2006). In this study, integrating physiographic maps, terrain derivatives, and remote sensing data proved effective in capturing the spatial heterogeneity of available N. The inclusion of high-resolution terrain and spectral data enabled the models to account for fine-scale variations caused by soil formation processes, erosion, and depositional dynamics, which are particularly relevant in complex landscapes like northeast India. The analysis highlights the central role of terrain attributes, followed by spectral and climatic variables, in predicting soil available nitrogen. TRI, LS-factor, MRRTF, and selected Landsat bands were consistently significant across different ML algorithms, emphasizing their utility in digital soil mapping.

3.5. *Predictive maps of N and quantified uncertainties*

The RF predicted N maps present a detailed spatial characterization of available nitrogen content and its associated uncertainty across the

study area. These maps integrate multiple model iterations to provide a robust representation of spatial variability, offering valuable insights into soil N distribution and reliability of model predictions. The mean predicted N maps (Fig. 9a and 9b) illustrate the average nitrogen concentration derived from 50 independent runs of the RF model, ensuring a statistically stable estimation of soil nitrogen across varying landscape features. The predicted N values range from 172 to 438 kg ha^{-1}, reflecting considerable heterogeneity in nitrogen availability that corresponds to the underlying geomorphic and soil-forming processes of the region.Spatially, higher concentrations of available nitrogen were observed in the older alluvial plains, where finer-textured soils, organic matter accumulation, and long-term cultivation likely contribute to nutrient enrichment. In contrast, the young alluvial plains, characterized by relatively coarser materials and frequent depositional activity, exhibited lower nitrogen levels. This spatial differentiation aligns with the soil–landscape relationships in alluvial systems, where variations in texture, moisture regime, and organic inputs influence nitrogen retention and mineralization. The RF model effectively captured these subtle environmental gradients, demonstrating its strength in modeling nonlinear interactions between nitrogen content and multiple predictor variables such as terrain indices, vegetation indices, and climatic covariates. Fig. 8c and 8d display the corresponding uncertainty maps, represented as the standard deviation of predictions across 50 RF iterations. The uncertainty values ranged from 101 to 111 kg ha^{-1}, indicating moderate variability in model stability across the study area. The relatively narrow range of standard deviation values suggests that the RF model performed consistently across most locations, with limited sensitivity to random sampling effects. However, slightly elevated uncertainty zones were evident in transitional areas between geomorphic units and along drainage channels. These areas often exhibit high microvariability in soil properties and land use, which can reduce prediction confidence. The inclusion of uncertainty mapping is particularly valuable for spatial soil assessments, as it provides a quantitative measure of confidence associated with model outputs. Such maps assist in identifying regions where additional field sampling or data refinement may be necessary. Overall, the RF-predicted nitrogen maps, coupled with

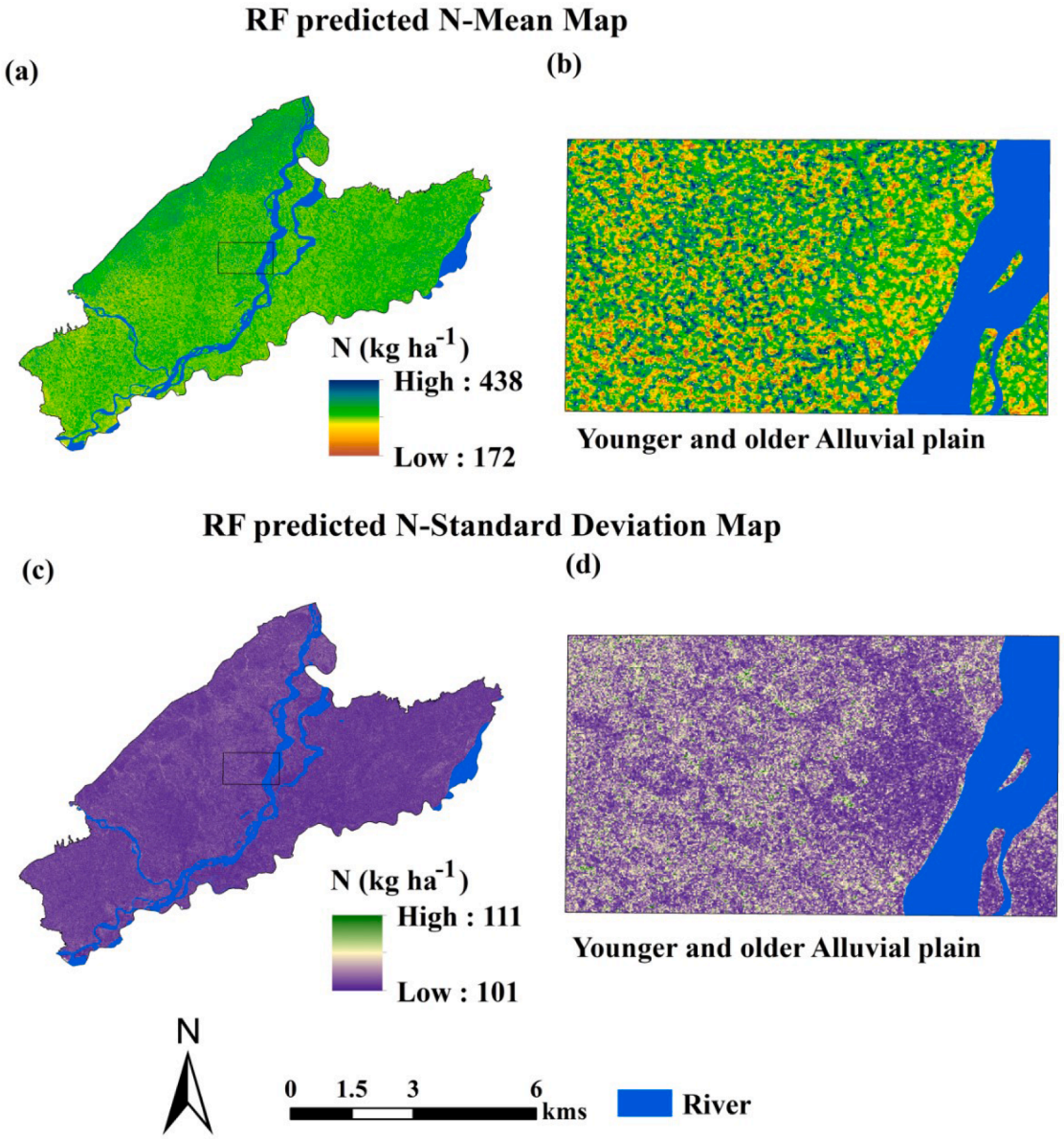


Fig. 9. Digital maps of nitrogen (N) were produced by best model (random forest): (a) mean N map over the study area resulting from the prediction among 50 iterations; (b) enlarged view of mean N in young and older alluvial plain area of study location; (c) uncertainty map of N over the study area resulting from the standard deviation of 50 iterations and (d) enlarged view of standard deviation in young and older alluvial plain area of study area.

uncertainty assessment, offer an integrated view of both soil nutrient distribution and prediction reliability. This approach enhances the interpretability of digital soil mapping outputs and supports informed decision-making in precision agriculture and sustainable nutrient management.

The prediction's low variability was observed near the bank region or the place with the lower elevation between the land areas. This might be due to the N solubility and loss from the soil to the water under flooding conditions. The spatial distribution and respective variation are almost homogenously distributed over the mainland. The prediction suggested that the variability in the soil available N is highest within the small land coverage. This might be due to the small land holding and the undulated land condition. The study area's eastern sections, which have soils prone to severe erosion and agricultural cultivation, have low values of N. In contrast, the western region, dominated by forest and thick vegetation, has significant values of N in all models. These spatial patterns of N seem acceptable. Low-elevation cultivable fields in the upper Brahmaputra valley seem more doubtful than high-elevation cultivable grounds. This makes sense, given the differences in the management approaches employed in the areas with higher levels of intensive farming (Jena et al. 2022). These downstream-farmed regions' geographic variation in N content levels may have been caused by management factors that the variables included in these prediction models may not have sufficiently accounted for. Management may have less influence since the terrain features have a better grip on the high-elevation areas and are not as intensively farmed. The uncertainty analysis can be used to confirm the trend of the ML algorithms' ability for predicting N (Dar and Parthasarathy, 2023). The lowest uncertainty was observed in the northeastern and central regions, and the highest uncertainty in the southeast and southwest regions might be due to the higher and lower sampling points, respectively. This might be attributed to the uncertainty framework. First, the expected uncertainty is a function of sample density. Our analysis showed that the area with high elevation, few sampling sites, and widely scattered fields had more uncertainty. This arose from the demand for further information on the relationships between environmental variables and soil. While this happens, higher accuracy in entrapment of the environmental spatial variability is feasible and subsequently significant reduction in sampling site deviation. This is especially true in areas with notable changes in the environment and soil characteristics (Dharumarajan et al. 2019; Liang et al. 2019). Second, uncertainty is influenced by the quantity and quality of environmental factors. In this investigation, maximum efforts have been attempted to collect the environmental data towards forecasting and predicting the nitrogen. Enhancing the sample density, identifying and recording valuable variables, and optimising spatial modelling tools to minimise uncertainty is recommended. To reduce the sampling and modelling uncertainty the high-quality sample data and environmental factors is used to improve the accuracy and dependability of the data products, especially in regions, where, soil and landscape change rapidly.

3.6. Area of applicability (AoA) of spatial prediction model

The Area of Applicability (AoA) is a diagnostic tool used to identify where a spatial prediction model can be considered reliable based on the similarity between the conditions of the prediction area and those used for model training. In spatial modeling, especially with machine learning, models often perform well only within the environmental range represented by their training data. The AoA quantifies this by computing a dissimilarity index (DI) that measures how similar each prediction location's covariate values are to the multivariate distribution of the training data. Areas with low DI values fall within the AoA, indicating that predictions there are supported by known conditions, while areas with high DI values are outside the AoA, where extrapolation occurs.For available N prediction, the DI map visualizes the level of dissimilarity between the local environment and the conditions under which the model was trained (Fig. 10). Areas with a high DI indicate locations where the input data differs significantly from the training data, suggesting the model may be extrapolating and its predictions for N levels are potentially unreliable. Complementing this, the AoAmap directly classifies each location. It assigns a value of '1′ (Applicable) where the model is deemed reliable and a value of '0′ (Not Applicable) where it is not. This binary classification is likely derived from a specific DI threshold.By comparing these maps; one can identify regions where the model's predictions should be used with caution or not at all. Areas marked as 'Not Applicable' (0) are expected to align closely with zones of high dissimilarity (DI $\geq$ 4). This combined visualization provides essential guidance, ensuring that the available N predictions are applied judiciously. It helps end-users identify and avoid areas of high uncertainty, thereby supporting more confident and accurate decision-making in soil nutrient management.

4. Conclusions and recommendations

This study examined the available N distribution in northeastern India's upper Brahmaputra basin. The available N in the research region was predicted by analysing it using five machine learning algorithms: RF, Cubist, SVM, XGBoost, and KNN. It computed the geographical distribution patterns of N in the upper Brahmaputra region of northeastern India. The spatial distribution of N was found in this study using information, environmental factors, and optimal models. Our results suggest that:

1. Integrating remote sensing data with machine learning models provided reliable predictions of available nitrogen (N) in the upper Brahmaputra valley.
2. Among the models tested, Random Forest (RF) demonstrated reasonable predictive performance compared to XGBoost, SVM, Cubist, and KNN, as indicated by its higher R^2 and lower RMSE values.
3. Under random cross-validation, RF consistently outperformed other models ($R^2c = 0.96$, RMSEc = 40.69 kg ha^{-1} in calibration and $R^2v = 0.35$, RMSEv = 70.44 kg ha^{-1} in validation). In contrast, spatial cross-validation resulted in markedly lower accuracy across all models, with R^2 values varied between 0.02 and 0.27.
4. The most influential predictor variables in the RF model were topographic roughness index (TRI), Landsat Band 2, Band 7, multi-resolution ridge top flatness (MRRTF), Band 4, mean annual temperature (MAT), and topographic position index (TPI). In the Cubist model, channel network base level (CNBL), slope, MRRTF, LS factor, elevation, and Band 7 were the key contributors. For SVM and KNN models, TRI, LS factor, convergence index (CI), MRRTF, and aspect emerged as the primary controlling factors.
5. Predicted N values varied between 172 and 438 kg ha^{-1}, while uncertainty, expressed as standard deviation, ranged from 101 to 111 kg ha^{-1}.The Dissimilarity Index (DI) and Area of Applicability (AOA) maps were used to delineate areas where model predictions are most reliable.
6. Elevated uncertainty observed in some areas of the upper Brahmaputra valley highlights the need to refine spatial modeling approaches. Limited ability to capture low variability may be linked to covariate selection. Nevertheless, these predictions offer valuable guidance for comprehensive soil quality monitoring and informed decision-making in the region.

Nitrogen is a key nutrient for plant growth but is highly susceptible to losses from soil erosion and other environmental processes. These dynamics must be considered when managing nitrogen availability at the farm level. The increasing demand for spatially explicit information on nutrients has been addressed through digital soil mapping (DSM) techniques. In this study, DSM was integrated with high-resolution environmental covariates to generate detailed regional predictions of

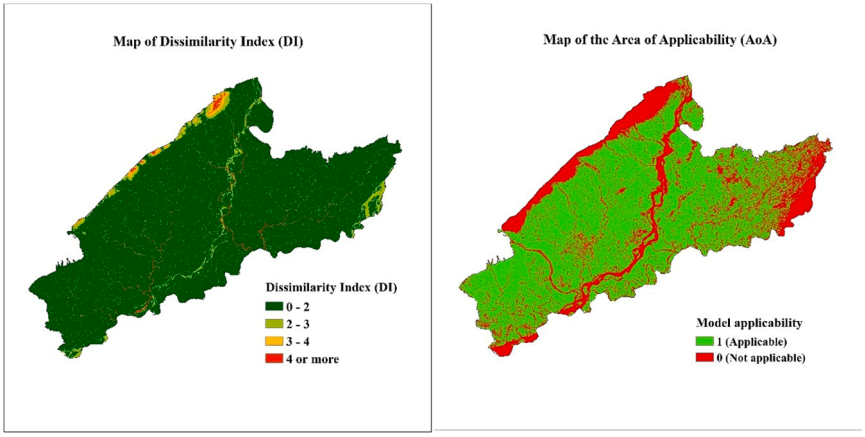


Fig. 10. Identifying regions for reliable model application for available N prediction.

soil available nitrogen in northeast India. This approach improved the accuracy of N distribution estimates, capturing spatial variability at both regional and local scales. The resulting nitrogen maps highlight potential sources of uncertainty, providing guidance for future high-resolution DSM studies and for monitoring soil quality changes under intensive cropping. Fine-resolution maps of available nitrogen are valuable for soil scientists, environmental managers, and land-use planners seeking to implement site-specific nutrient management strategies. Given the pronounced heterogeneity of agro-ecological zones across northeast India, the methodology demonstrated in this study can serve as a template for mapping available nitrogen in other regions. Overall, the findings underscore the importance of combining DSM with environmental covariates to enhance nutrient management planning. High-resolution nitrogen maps can support sustainable agriculture by informing precise fertilization strategies, reducing environmental impacts, and enabling effective decision-making for resource allocation and land management.

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CRediT authorship contribution statement

Amit Kumar: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Pravash Chandra Moharana:** Writing – review & editing, Validation, Software, Methodology, Formal analysis. **Roomesh Kumar Jena:** Software, Methodology, Formal analysis. **Sandeep Kumar Malyan:** Writing – original draft. **Ram Kishor Fagodiya:** Writing – review & editing. **Aftab Ahmad Shabnam:** Visualization, Supervision. **S.N Vinodakumar:** Data curation. **Dharmendra Kumar Jigyasu:** Data curation. **Azad Gull:** Writing – review & editing, Writing – original draft. **Kasthala Mary VijayaKumari:** Visualization, Supervision, Project administration. **Gandhi Doss Subramaniam:** Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known conflict financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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